Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Technical Architecture:

User (Researcher/Policymaker) → Web Interface → Flask Application → Data Preprocessing → Machine Learning Model → HDI Prediction → Results Dashboard ↘ Dataset (CSV)

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for entering HDI indicators	HTML, CSS, Bootstrap
2	Application Logic-1	Input validation & preprocessing	Python, Flask
3	Application Logic-2	Prediction workflow	Scikit-learn
4	Application Logic-3	Model evaluation	Pandas, NumPy
5	Database	Dataset storage	CSV / SQLite
6	Cloud Database	Optional cloud storage	Firebase / MongoDB Atlas
7	File Storage	Store datasets & models	Local File System
8	External API-1	Optional HDI data source	World Bank API
9	External API-2	Country information	REST Countries API
10	Machine Learning Model	HDI Classification	Random Forest Classifier
11	Infrastructure (Server / Cloud)	Deploy application	Local / Render / PythonAnywhere

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used	Flask, Scikit-learn, Pandas
2	Security Implementations	Input validation & secure deployment	HTTPS, Input Validation
3	Scalable	Modular ML	Flask + REST

	Architecture	architecture	
4	Availability	Accessible through web browser	Cloud Deployment
5	Performance	Fast prediction using trained model	Scikit-learn